

Creating Custom Tags in Jakarta Server Pages

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Introduction

Custom tags are reusable components that allow developers to add application-specific elements to Jakarta Server Pages (JSPs). Instead of repeating Java code or complex logic directly in a JSP page, a developer can create a tag with a clear name, such as `<app:greeting>` or `<app:message>`. When the page is processed, the web container connects the tag to a Java handler class or tag file that performs the requested task. Custom tags can make JSP applications easier to read, organize, and maintain. They are especially useful when the same presentation appears on several pages. However, they also require additional setup and can complicate an application when used incorrectly. Developers should understand both the benefits and limitations of custom tags before adding them to a project.

What Custom Tags Are

A custom tag is a developer-created JSP element that performs a specific function. It resembles an HTML or XML tag but is defined by the application rather than the browser. For example, a developer could create the following tag:

```
<app:greeting name="Jose" />
```

The tag could display a personalized greeting without requiring the JSP page to contain Java code. A custom tag can also contain body content:

```
<app:panel title="Account Information">
```

Customer information appears here.

```
</app:panel>
```

Custom tags are usually grouped into tag libraries. A tag library contains related tags that may be reused across an application. Custom tags allow developers to extend the JSP language and place reusable functionality behind simpler elements. There are two common methods of creating a custom tag. The first is to write a Java handler class. A handler class is useful when the tag must perform more

advanced Java processing. The second method is to create a tag file with the .tag extension. Tag files use JSP-like syntax and are generally easier to use when the component primarily produces HTML content.

Advantages of Custom Tags

One of the greatest advantages of custom tags is code reuse. A developer can create a feature once and use it on several JSP pages. For example, an application may repeatedly display status messages, prices, customer records, navigation menus, or permission-based content. Without a custom tag, similar code may need to be copied into each page. Duplication makes maintenance more difficult because every copy must be updated separately. Custom tags also improve readability. A short-named tag usually communicates its purpose better than a long JavaScript snippet. The example below demonstrates this.

```
<app:formatPrice amount="${product.price}" />
```

The purpose of the tag is easy to understand. The page does not need to include Java objects, formatting instructions, or output statements. Those details are hidden inside the tag implementation.

Another benefit is better separation of concerns. JSP pages should mainly control how information is displayed. Business rules and complicated Java logic should be handled in JavaBeans, service classes, controllers, or other application components. Custom tags help remove reusable presentation processing from the page itself. The Jakarta Server Pages specification encourages the use of reusable components and tag libraries instead of filling pages with JavaScriptlets (Oracle, 2007).

Custom tags can also accept attributes. Attributes make a tag flexible to work with different values. A greeting tag could accept a person's name, while a message tag could accept a title and description. Some tags can process their body content or use Expression Language values from the JSP page. Maintenance may become easier because the tag's behavior is stored in one location. If a formatting rule changes, the developer can update the handler rather than editing every page that uses it. This reduces inconsistent output and lowers the chance of forgetting one.

Disadvantages of Custom Tags

Although custom tags have useful advantages, they can become complicated. A Java-based tag may require a handler class, a package structure, a Tag Library Descriptor file, a compiled class, a JSP directive, and the correct deployment location. A mistake in the URI, class name, attribute definition, or directory structure may prevent the JSP page from compiling. Troubleshooting can also be more difficult for beginners. The JSP page may contain only a single short custom tag, but the actual error may be in the Java handler or descriptor file. Developers need to understand how the JSP container connects all parts.

Another disadvantage is that a custom tag may duplicate features already available through the Jakarta Standard Tag Library, or JSTL. JSTL already provides tags for conditions, loops, formatting, URLs, internationalization, and other common tasks. The Jakarta Tags specification describes elements that developers can use across different JSP applications (Oracle, 2007). Creating a custom loop or conditional tag would add unnecessary steps when a standard solution already exists. Custom tags may also hide too much logic. A tag that performs database updates, controls major business decisions, or contains a large workflow can make the application difficult to understand. Custom tags are usually most effective for presentation tasks. Database access and business processing should remain in appropriate Java classes rather than being hidden inside page tags.

There is also a learning cost for team members. Standard JSP and JSTL tags are widely documented, but a custom tag library is unique to an application. New developers must learn its attributes, behavior, and expected use.

Opinion

Custom tags are valuable when a JSP application contains repeated behavior that cannot be handled clearly with JSTL, Expression Language, JSP includes, or JavaBeans. A well-named custom tag can make a page easier to read and reduce duplicated code. However, I would not use custom tags for

every task. First, check whether JSTL already provides the needed feature. For a component that mostly contains HTML, I would choose a tag file because it is easier to create and maintain. I would use a Java handler only when the component needs more substantial Java processing. I would avoid a custom tag when it is used on only one page or when its setup is more complicated than the code it replaces. I would also avoid placing database updates or important business logic inside tags. Custom tags are most useful when they provide a clear, reusable presentation feature without hiding the application's main processing.

Conclusion

Custom tags extend JSP by allowing developers to create reusable application-specific elements. Their main benefits include code reuse, improved readability, consistent presentation, and easier maintenance. Their disadvantages include additional configuration, more difficult troubleshooting, and the possibility of duplicating standard JSTL features. When used selectively, custom tags can improve the structure of a JSP application. When used unnecessarily or for complex business processing, they may create more problems than they solve.

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